

OceanBed

Seeing 1000 m beneath the ocean surface — from satellites alone

- **Problem Statement ID – SIH26066**
- **Problem Statement Title – OceanEmbed – Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations**
- **Theme – Space Technology**
- **PS Category – Software**
- **Team ID – [fill in]**
- **Team Name (Registered on portal) – [fill in]**



Your
Team
Name

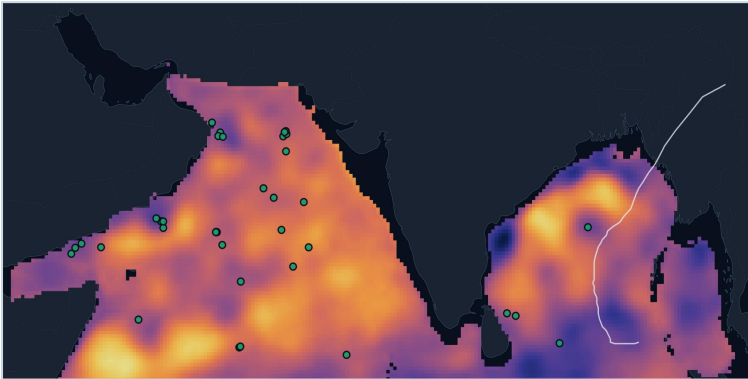
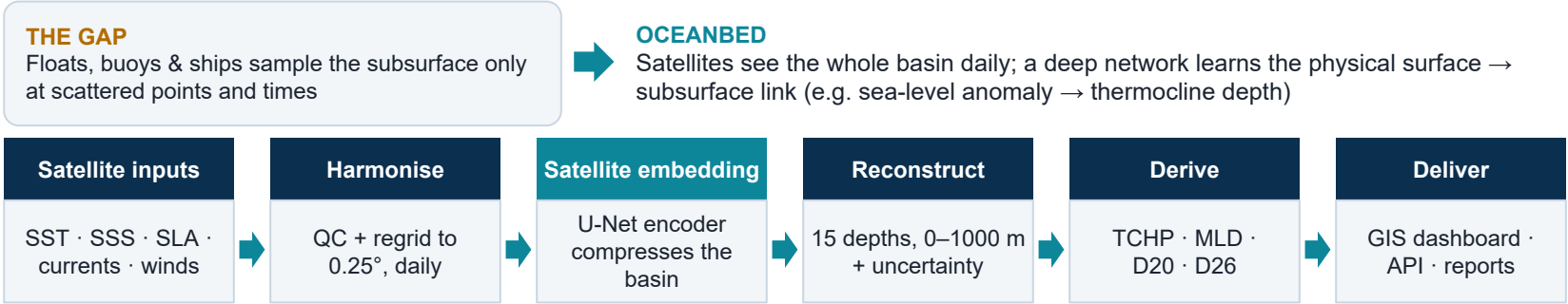
OceanBed

Daily 0–1000 m ocean temperature for the North Indian Ocean — from satellites alone

Proposed Solution Detailed explanation of the proposed solution

OceanBed reconstructs ocean temperature at **15 depths (0–1000 m)**, **every day**, on a **0.25° grid** over the Bay of Bengal & Arabian Sea — using only **5 satellite surface fields** — and turns it into cyclone-heat and mixed-layer products for ocean analysts and forecasters.

How it addresses the problem



Working prototype: reconstructed temperature at 100 m on 11 May 2023 (held-out year) — dark purple ≈ 17 °C to yellow ≈ 28 °C; green dots = Argo floats; white line = Cyclone Mocha track (IBTrACS)

INNOVATION AND UNIQUENESS OF THE SOLUTION — why OceanBed is different

USP 01 Validated beyond training data

Scored against real Argo floats from a year it never saw — next to the training product's own error ceiling and an independent EN4 cross-check.

2,639 held-out Argo profiles (2023)

USP 02 Calibrated to the real ocean

Every profile carries an error bar calibrated on 2022 floats that holds on 2023 — analysts can see when not to trust a value.

70% of Argo values within $\pm 1\sigma$ (ideal 68%)

USP 03 Reconstruction → decisions

Daily cyclone heat potential, mixed-layer and isotherm depths at 0.25°, replayed on real IBTrACS cyclone tracks.

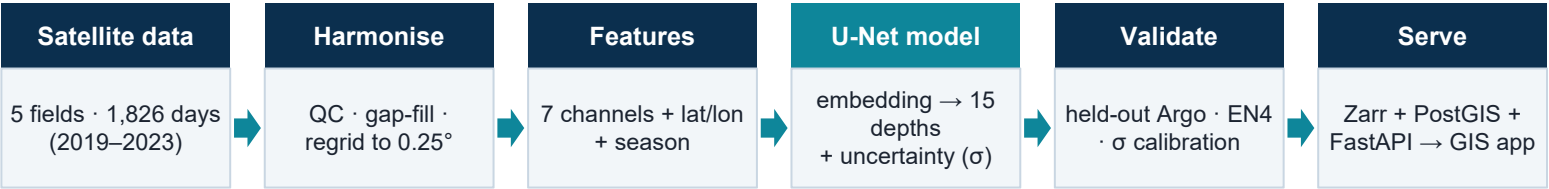
TCHP · MLD D20 · D26 — every day, 2019–2023

Satellite surface fields → validated, uncertainty-aware 3-D ocean temperature → INCOIS decision products

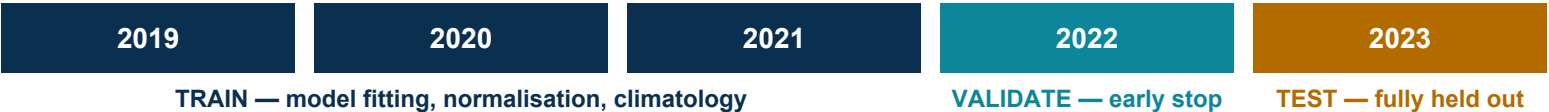
Technologies to be used

Satellite data	NOAA OISST (SST) · SMAP + SMOS (SSS) · NOAA altimetry (SLA, currents) · NCEI Seawinds
Target & validation	HYCOM 1/12° analysis (GLORYS adapter built) · Argo via argopy · Met Office EN4
ML / AI	PyTorch U-Net (satellite embedding) · LightGBM baseline
Data engineering	Python · xarray · Zarr (daily cubes) · NumPy regridding
Geospatial / DB	PostgreSQL + PostGIS (spatial indexes, nearest-float queries)
Backend	FastAPI — typed REST API, PDF/CSV reports
Frontend / GIS	Next.js · MapLibre GL · deck.gl · Recharts
Engineering	Docker · GitHub Actions CI · 44 automated tests · CPU-only

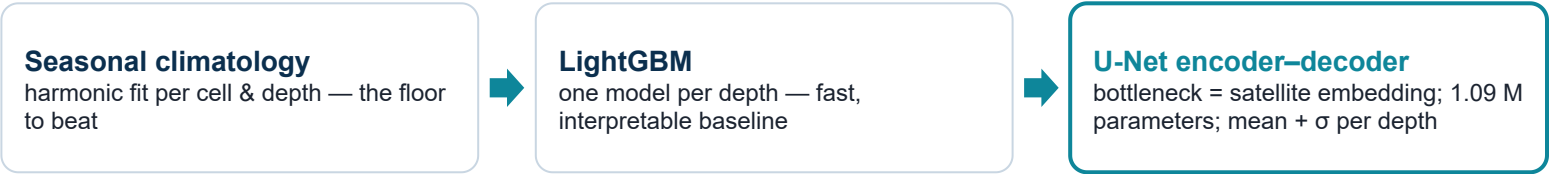
Methodology and process for implementation



Honest evaluation — split by whole years, never random days



Models, compared side by side — the deep model has to earn its place



VALIDATION — against the real ocean, not just the training data

15,018

Argo profiles, QC flags 1/2

2,639

in the held-out 2023 test

per depth

RMSE · bias · r · skill vs
climatology

EN4

independent Met Office cross-
check

Analysis of the feasibility of the idea
 working prototype, validated on the held-out year 2023

2,639

independent Argo profiles from 2023 — a year the model never trained on

0.89 °C

mean temperature error (RMSE, 15 depths) vs 1.10 °C for climatology

1.31 °C

error at 100 m (thermocline) vs 1.80 °C for climatology

70%

of Argo values within $\pm 1\sigma$ after calibration (ideal 68%; raw 41%)

FEASIBLE BECAUSE

Data HIGH	Every input, target and validation source used is open; Copernicus/ERA5 need only free sign-up
Technical HIGH	Built and running end to end: ingestion → models → API → dashboard
Compute HIGH	U-Net (1.09 M params) trained on a laptop CPU — no GPU
Cost ₹0	No paid data or services used

IMPLEMENTED
 pipeline · models · validation · API · dashboard · Docker

PLANNED hosted deployment · re-run on Copernicus/GLORYS with credentials

Potential challenges and risks

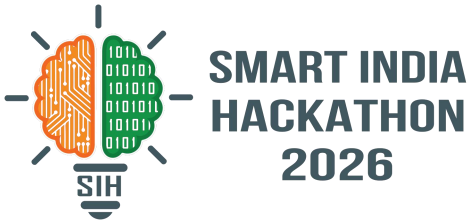
Training target (a reanalysis) assimilates Argo → validation not fully independent
Raw model uncertainty was over-confident (41% of values within $\pm 1\sigma$)
Satellite salinity is gappy; removing it showed no measurable change in skill
GLORYS / Copernicus data need an account
Thermocline (20–150 m) is the hardest layer to reconstruct

Strategies for overcoming these challenges

Hold out a whole year; show the target product's own error as the ceiling (0.79 °C); cross-check against EN4
Per-depth calibration fitted on 2022 floats, verified on 2023 → 70% within $\pm 1\sigma$, 94% within $\pm 2\sigma$
Merged SMAP + bias-corrected SMOS (absent days 209 → 1); re-test with Copernicus SSS
Credentialed adapters already built — switching source is a config change, no code change
Report error per depth, never one number; the uncertainty band widens where skill drops

Your
Team
Name

IMPACT AND BENEFITS



Potential impact on the target audience

Ocean analysts & forecasters
INCOIS-style

Cyclone forecasters
heat available to storms

Fisheries-advisory analysts
mixed layer & thermocline

Ocean researchers
daily 3-D record

Decision support — not a replacement for operational ocean models

Benefits of the solution social · economic · environmental · operational

SOCIAL

Basin-wide cyclone heat potential (TCHP) and mixed-layer maps for every day and every 0.25° cell of the domain.

ECONOMIC

Built only on free satellite data; the model trains on a CPU; products are precomputed, so viewing needs no heavy compute.

ENVIRONMENTAL / CLIMATE

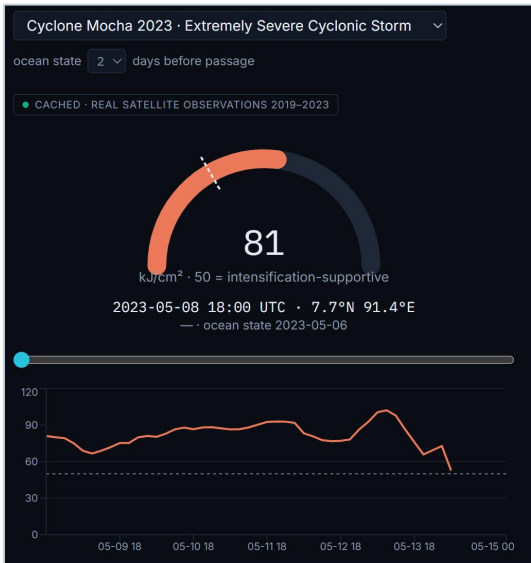
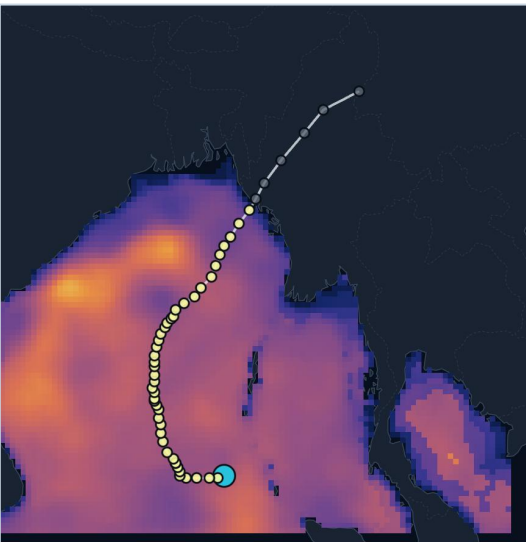
A continuous daily record of upper-ocean heat and stratification for the North Indian Ocean (2019–2023).

OPERATIONAL

Each profile ships with a calibrated error bar and the nearest independent Argo float — trust is visible, not assumed.

KEY OUTCOME full-basin subsurface view · daily · 0–1000 m · with measured confidence

Working prototype real event, held-out year



Cyclone Fuel Gauge — reconstructed TCHP along Cyclone Mocha's real IBTrACS track (May 2023, ocean state 2 days before passage): 52.7–102 kJ/cm² over the ocean, above the 50 kJ/cm² level commonly linked to intensification. Prototype output, not an operational forecast.

Click any cell
full 0–1000 m profile + σ

Argo overlay
nearest independent float

Export
PDF / CSV report per point

Details / Links of the reference and research work

#	Reference	Organisation	What it supports	Link
1	SIH26066 OceanEmbed problem statement	SIH · MoES / INCOIS	Scope, domain, 15 standard depths	sih.gov.in
2	INCOIS Tropical Cyclone Heat Potential service	INCOIS	Operational relevance of TCHP	incois.gov.in/site/services/tchp.jsp
3	GLORYS12V1 global ocean reanalysis (PS-named target)	Copernicus Marine	Training-target specification	doi.org/10.48670/moi-00021
4	Argo float profiles via argopy	Argo programme · Euro-Argo	Independent validation	argopy.readthedocs.io
5	NOAA OISST v2.1 · NCEI Blended Seawinds	NOAA NCEI	Satellite inputs used	ncei.noaa.gov
6	EN4 quality-controlled ocean analysis	UK Met Office Hadley Centre	Independent cross-check	metoffice.gov.uk/hadobs/en4
7	Bay of Bengal barrier layer (Vinayachandran et al., 2002)	JGR Oceans · AGU	Why salinity is an input	doi.org/10.1029/2001JC000831
8	Subsurface temperature reconstruction, ConvLSTM vs LightGBM (2022)	MDPI Remote Sensing	Model choice; thermocline is hardest	doi.org/10.3390/rs14133198
9	IBTrACS v04r01 best-track archive	NOAA NCEI	Real cyclone tracks	ncei.noaa.gov/products/international-best-track-archive

Prototype source code & full research artifact: github.com/saad-46/oceanembed · all reported metrics are generated by the pipeline (docs/RESULTS.md)